Recap Checkpoint 6 — Full Year 12 Cumulative (Pre-Mock): 1.1–1.4 + 2.1 + 2.2.1

Name: ______________________ Date: __________ Mark: ______ / 46

OCR H446 cumulative recap — mixes every topic taught so far. Show all working.

Questions

Q1 [3] (AO1). *(1.1 — Processors)* State **three** factors that affect the **performance** of a processor, and for each give a brief explanation of its effect.

Q2 [4] (AO2). *(1.4 — Number bases)* Show all working.

(a) Convert denary **181** to **8-bit binary**. **[1]**

(b) Convert **181** to **hexadecimal**. **[2]**

(c) Convert binary `1010 0110` to denary. **[1]**

Q3 [4] (AO2). *(1.4 — Two's complement)* Using **8-bit two's complement**, calculate **−40 + 25**.

(a) Show 40 and 25 in 8-bit binary. **[1]**

(b) Form the two's complement representation of −40. **[1]**

(c) Perform the addition and give the final **denary** result. **[2]**

Q4 [3] (AO2). *(1.4 — Boolean algebra / floating point)*

(a) State why floating-point numbers are **normalised**. **[1]**

(b) Simplify **A ∨ A∧B** and name the law used. **[2]**

Q5 [4] (AO1). *(1.3 — Networks)*

(a) Explain the difference between **circuit switching** and **packet switching**. **[2]**

(b) State **two** items of data added to a packet's **header**. **[2]**

Q6 [3] (AO1). *(1.2 — Operating systems)*

(a) State what is meant by **scheduling** and name **one** scheduling algorithm. **[2]**

(b) State the purpose of a **device driver**. **[1]**

Q7 [4] (AO2). *(2.1 / 2.2.1 — Computational thinking + programming)*

(a) State the difference between a **procedure** and a **function**. **[1]**

(b) Explain the difference between **passing a parameter by value** and **by reference**. **[2]**

(c) State **one** advantage of using **local** variables over **global** variables. **[1]**

Q8 [5] (AO2). *(1.4 — Data structures trace)* A **stack** is initially empty. The following operations are carried out in order: `push(7)`, `push(2)`, `push(9)`, `pop()`, `push(4)`, `pop()`, `pop()`.

(a) State the value returned by **each** of the three `pop()` calls, in order. **[2]**

(b) State the contents of the stack after all seven operations. **[1]**

(c) The same seven operations are repeated using a **queue** (`enqueue` in place of `push`, `dequeue` in place of `pop`). State the values returned by the three `dequeue()` calls, and name the access order each structure demonstrates. **[2]**

Q9 [4] (AO1/AO2). *(2.1 — Computational thinking)*

(a) State what is meant by **abstraction** and give **one** example of its use in software. **[2]**

(b) **Caching** is often described as an example of *thinking ahead*. State what caching is and give **one** benefit it provides. **[2]**

Q10 [6] (AO1/AO2). *(1.4 — Boolean logic: adders)*

(a) Complete the **Sum** and **Carry** columns of the truth table for a **half adder**. **[2]**

A	B	Sum	Carry
0	0		
0	1		
1	0		
1	1		

(b) Name the **logic gate** that produces the Sum output and the logic gate that produces the Carry output. **[2]**

(c) State **one** way a **full adder** differs from a half adder, and explain why this difference allows full adders to be chained together to add multi-bit binary numbers. **[2]**

Q11 [6] (AO3). *(Synoptic extended response)* A hospital is replacing a paper appointment system with networked software that stores patient data, lets staff search records, and sends automated reminders.

In an extended response, discuss the design considering: **(i)** how **abstraction and decomposition** help manage the build, **(ii)** an appropriate **data structure** for fast record lookup (justify your choice), and **(iii) one security** consideration for transmitting and storing patient data. **[6]**